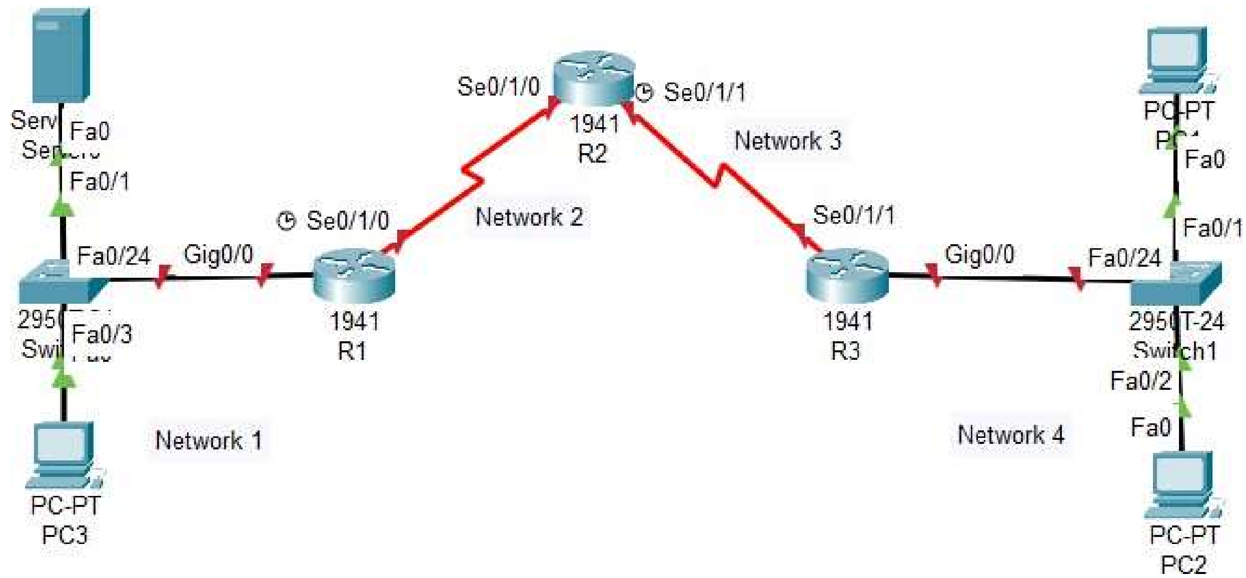
CCS6224
NETWORK SECURITY
Lab 4
(Graded Lab Exercise 2)

<u>Practical Lab: (a) Configuring Access Control Lists (ACLs) to mitigate network attacks</u>

1. Open the Packet Tracer and then construct the following network topology in the logical workspace.



Configure the network devices with the following basic configurations. **(1 mark)**

a) IP address

Network	IP Address
1	192.168.1.0/24 **(Graded submission: 192.StudentDayOfBirth.StudentMonthofBirth.0/24)**
2	172.16.1.0/30 **(Graded submission: 172.StudentDayOfBirth.StudentMonthofBirth.16/30)**
3	172.16.2.0/30 **(Graded submission: 172.StudentDayOfBirth.StudentMonthofBirth+100.8/30)**
4	192.168.3.0/24 **(Graded submission: 192.StudentDayOfBirth.StudentMonthofBirth+100.0/24)**

b) Configure OSPF as routing protocol

2. Configure and encrypt all passwords in R1, R2 and R3 **(1 mark)**

- Set password to minimum length = 8
- Set enable secret `cisco123` **(use ciscoStudentID for graded submission)**
- Set console password `ciscoconpw` **(use StudentIDconpw for submission)**
- Set telnet password `ciscovtypw` **(use StudentIDvtypw for submission)**
- Encrypt all clear text passwords in running-config file

<u>If your student ID is 1231302175, then your telnet password is 1231302175vtypw</u>
<u>If your student ID is 241UC240QK, then your telnet password is 241UC240QKvtypw</u>

3. Configure SSH server on router R2 **(2 mark)**

- Set fci_network.com as the domain name in R2
- Configure a user `admin` with the highest privilege level, set the password to be `ciscoadmin`

(For graded submission, use StudentIDadmin as username and StudentIDpass as password – replace with your own MMU Student ID)

- Configure the vty lines to only accept SSH connections
- Erase existing key pairs on the router
- Create the RSA encryption key pair for the router, with 1024 for the number of modulus bits
- Set the IP SSH to version 2, verify the SSH configuration
- Verify SSH connection from PC2 and PC3 to R2

4. Configure Secure Access Control to R2 using ACLs **(2 marks)**

a) On each router, create an ACL named VTY-Control to block all remote access to the router except PC2 and PC3
b) Apply the created ACL on the vty line of the router
c) Establish SSH connections to R2 from PC2 and PC3, successful or fail?
d) Establish SSH connections to R2 from PC1 and Server0, successful or fail?

5. Configure access control to the Server0 using extended ACL

a) Create an extended ACL named Server-Control on R1 with the below rules:
   - Permit hosts from any networks to access DNS, SMTP, HTTPS services on Server0
   - Permit only hosts from network 1 to access FTP service at Server0, block all outside hosts
b) Which interface in R1 do you think is most appropriate to apply this extended ACL to? Explain your answer
c) Apply the extended ACL to the specified interface
d) Verify that PC1 and PC2 can only access web service at Server0 via HTTPS, but not HTTP **(2 marks)**
e) Verify that PC1 and PC2 cannot access FTP service at Server0, only PC3 can access to Server0 via FTP **(2 marks)**

6. Save your configuration in the pkt file.

Note: For graded lab exercise submission, follow the instructions posted on eBwise on this matter.